

MAT 310, Handout 9: Hypergeometric Distribution (3.7)

Problem framework: Suppose you have N balls, r of which are red, while the other $N - r$ are black. Select a sample of size n . What is the probability that your sample includes y red balls?

Hypergeometric RV: Let $Y \sim \text{Hypergeometric}(N, n, r)$ be the number of successes from a sample of size n out of a population of size N with r successes. The distribution of Y is given by:

$$p(y) =$$

for $y = 0, 1, \dots, n$ subject to the restrictions $y \leq r$ and $n - y \leq N - r$.

Remark: The Hypergeometric distribution is similar in spirit to the Binomial distribution, except with a finite population and a finite number of successes to select. In fact,

$$\lim_{N \rightarrow \infty, r \rightarrow \infty, r/N=p} \frac{\binom{r}{y} \binom{N-r}{n-y}}{\binom{N}{n}} =$$

Example 1. Consider a 20-horse race. If you randomly select 5 horses to bet, find the probability that you selected:

(a) each of the top 3.

(b) at least one of the top 3.

Expected Value and Variance of $Y \sim \text{Hypergeometric}(N, n, r)$

$$E(Y) =$$

$$V(Y) =$$

Remark: As $N \rightarrow \infty$, the expected value and variance of the Hypergeometric distribution converge to the mean and variance of the Binomial distribution.

Example 2. Consider Example 1. Find the expected value and variance of the number of horses in the top 3 that you randomly select.

Example 3. In Keno Pick 4, there are 80 numbers, and you pick 4 of them. 20 of the 80 numbers are winners and flash on the screen, and the payoff for a \$1 bet is as follows:

# matches	payoff
0	\$0
1	\$0
2	\$1
3	\$5
4	\$55

Find the expected payoff for a \$1 bet.